

Ahmed Mohamed Salah Mwafy

6th of October City, Giza, Egypt

+20 1204943439

ahmedmwafy41@gmail.com

[LinkedIn](#)

[GitHub](#)

Career Objective

Motivated Mechatronics Engineering student seeking an opportunity in robotics and artificial intelligence to apply my experience in ROS 2, autonomous systems, motion planning, control, and robotic system integration. I aim to develop advanced skills in perception, planning, and intelligent robotic systems while contributing to innovative projects and growing as a robotics and AI engineer.

Education

(B.S.) Engineering, Mechatronics, University of Mansoura (2022 - 2027)

- GPA: 3.98 / 4.0 (Ranked 1st in my class)
- Status: Undergraduate

Work experience & Internships

Trainee, (Internship), Alexandria Shipyard, (August 2025 – September 2025)

- I assisted in mechanical and electrical maintenance of ship systems.
- I applied classic control concepts in ship automation and propulsion systems.
- I learned safety standards and quality procedures in shipbuilding operations. Achievements
- I built connections working with senior engineers.
- I enhanced my knowledge of safety and quality assurance in industrial environments.

Ros Developer, (Internship), Karthikesh Robotics, (June 2025 – August 2025)

- I developed ROS/ROS 2 nodes for motion planning and control systems.
- I tested robotic systems in Gazebo and RViz.
- I used camera for perception tasks such as environment understanding.
- I built a ROS based camera line follower robot and test it in Gazebo.

Extracurricular activities

Autonomous member, Motorsport, Mansoura, November 2024 - Present

- I worked on path planning and control systems for autonomous racing cars.
- I Applied algorithms such as Pure Pursuit, and PID for control system.
- I assisted in integrating sensors (camera, Lidar, IMU) for perception and localization.
- I coordinated with mechanical and electrical sub-teams for system integration.
- I delivered a training session on motion planning for university students.
- We secured a ranking in the F1TENTH ICRA competition qualifiers.

Technical Director, Momentum, Mansoura, January 2024 - Present

- I mentored students in multiple robotics and embedded systems projects, providing technical guidance and support.
- I served as Vice Embedded Head, assisting in system design and mentoring the embedded sub-team and presented multiple sessions including one in a technical workshop.
- I acted as Technical Director, coordinating technical work across AI, embedded, Hardware and mechanical teams.
- I delivered a session on project management for technical members.
- I served as Project Manager in Micro mouse competition.
- I served as Booth Head of technical presenters in November 2024.
- I participated in Plaza competition, contributing to line follower and robotic arm development.

Training Courses

Embedded Systems Basic Diploma – Eng. Amgad Samir, July 2024

- Learned Embedded C and AVR microcontrollers
- Studied microcontroller architecture and embedded systems building process

- Gained basics of RTOS and embedded software development
- Automation Basic Diploma, HA Consulting Group, September 2025
- Studied classical control systems and industrial automation fundamentals
 - Worked with PLC programming (ladder logic) and motor control systems
 - [Certificates](#)

Embedded Linux Diploma – Eng. Moatasem, January 2025

- Learned C++, Python, and Linux fundamentals
- Worked with Raspberry Pi and Embedded Linux basics
- Gained experience with Linux-based embedded development

MATLAB Onramp, MathWorks, August 2025

- I learned basics of programming, data analysis, visualization, and working with arrays and matrices
- [Certificate](#)

Nxplorers pro workshop, Shell, July 2025

- Focus on problem-solving, systems thinking, and sustainable innovation.
- [Certificate](#)

ROS workshop, Karthikesh Robotics, June 2025

- I worked on ROS-based robotic systems and simulation, including a camera-driven line follower project, with focus on system optimization and problem-solving.
- [Certificate](#)

TOT (Training of Trainers), Momentum, February 2024

- A professional development program that prepares participants to design, deliver, and evaluate training sessions.
- [Certificate](#)

Management, Momentum, July 2024

- I gained skills in leadership, team management, communication, and organizational development.
- [Certificate](#)

Skills

Programming & Software

- C, C++, Python
- Data Structures & Algorithms
- MATLAB & Simulink
- Linux

Control & Automation

- Classic Control Systems
- PLC Programming & Interfacing (Siemens, Schneider, Kinco)
- Ladder Diagram Programming
- Factory IO
- Siemens TIA Portal

Embedded Systems & Hardware

- Microcontroller Interfacing
- Sensor Interfacing
- Communication Protocols (UART, SPI, I2C)
- Electronics & Hardware Prototyping
- AVR Microcontrollers

Robotics & Simulation

- ROS 2 & Gazebo
- Proteus

Other Skills

- Project Management
 - Presentation Skills
-

Languages

- Arabic – Native
- English – B2

References are available upon request.

Date of birth	23/09/2004	Nationality	Egypt
Marital status	Single	Military status	Exempt – only son
Gender	Male		